

UNITS & MEASUREMENTS

MEASUREMENT OF MASS & TIME

MASS

- Unified atomic mass unit (amu) is used to measure mass of atoms & molecules

- 1 amu = $(1/12)^{\text{th}}$ mass of one C^{12} atom
- 1 amu = 1.66×10^{-27} kg
- Electron mass = 10^{-30} kg
- Earth mass : 10^{25} kg
- Observable Universe 10^{55} kg

TIME

- SI unit is second (based on caesium clock with an uncertainty less than 1 part in 10^{-13} ie, 3µs loss every year)
- Timespan of unstable particle: 10^{-24} s
- Age of universe: 10^{17} s

MEASUREMENT OF LENGTH

- Large distance is measured by parallax method

$$\text{Parallax angle} = \frac{\text{BASIS}}{\text{DISTANCE}} = \frac{b}{x}$$

- $1^\circ = 1.745 \times 10^{-2}$ rad
- $1' = 2.91 \times 10^4$ rad.
- $1'' = 4.85 \times 10^6$ rad.

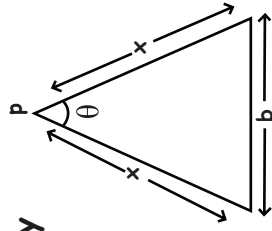
- For very small sizes, optical microscope, tunneling microscope, electron microscope are used.

- 1 AU = 1.496×10^{11} m
- 1 ly = 9.46×10^{15} m
- 1 parsec = 3.08×10^{16} m

- Size of proton: 10^{-15} m

- Radius Of Earth: 10^7 m

- Distance to Boundary Of Observable Universe: 10^{26} m

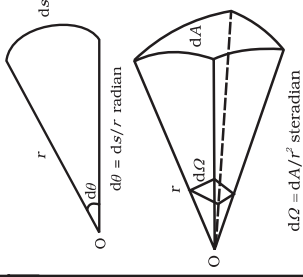


SI SYSTEM

7 Base units and 2 supplementary units

NO.	Base Units		
	Quantity	Unit	Symbol
1	Length	meter	m
2	Mass	kilogram	kg
3	Time	second	s
4	Temperature	kelvin	K
5	Electric current	ampere	A
6	Luminous intensity	candela	cd
7	Amount of substance	mole	mol

NO.	Supplementary Units		
	Quantity	Unit	Symbol
1	Plane angle	radian	rad
2	Solid angle	steradian	sr



What is the unit of permittivity of free space ϵ_0 ?

- colomb/newton-metre
- newton-metre² /colomb²
- colomb² /newton-metre²
- colomb² / (newton-metre)²

SIGNIFICANT FIGURES

The digits in a measured quantity which are reliable and confidence in our measurement + the digit which is uncertain.

RULES FOR SIGNIFICANT FIGURES

- All non-zero digits are significant. For example, 42.3 has three significant figures; 243.4 has four significant figures; and 24.123 has five significant figures.
- A zero becomes significant figure if it appears between two non-zero digits. For example, 5.03 has three significant figures; 5.604 has four significant figures; and 4.004 has four significant figures.
- Leading zeros or the zeros placed to the left of the number are never significant. For example, 0.543 has three significant figures; 0.045 has two significant figures; and 0.006 has one significant figure.
- Trailing zeros or the zeros placed to the right of the number are significant. For example, 4.330 has four significant figures; 433.00 has five significant figures; and 343.000 has six significant figures.
- In exponential notation, the numerical portion gives the number of significant figures. For example, 1.32×10^{-3} has three significant figures and 1.32×10^4 has three significant figures.

RULES FOR ROUNDING OF A MEASUREMENT

- If the digit to be dropped is less than 5, then the preceding digit is left unchanged. For example, $x = 7.82$ is rounded off to 7.8 and $x = 3.94$ is rounded off to 3.9.
- If the digit to be dropped is more than 5, then the preceding digit is raised by one. For example, $x = 6.87$ is rounded off to 6.9 and $x = 12.78$ is rounded off to 12.8.
- If the digit to be dropped is 5 followed by digits other than zero, then the preceding digit is raised by one. For example, $x = 16.351$ is rounded off to 16.4 and $x = 6.758$ is rounded off to 6.8.
- If the digit to be dropped is 5 or 5 followed by zeros, then the preceding digit, if it is even, is left unchanged. For example, $x = 3.250$ becomes 3.2 on rounding off and $x = 12.650$ becomes 12.6 on rounding off.
- If the digit to be dropped is 5 or 5 followed by zeros, then the preceding digit, if it is odd, is raised by one. For example, $x = 3.750$ is rounded off to 3.8, again $x = 16.150$ is rounded off to 16.2.

RULES FOR ROUNDING OF A MEASUREMENT

ADDITION & SUBTRACTION

In addition or subtraction, the final result should be reported to the same number of decimal places as that of the original number with minimum number of decimal places

$$\begin{array}{r} 3.1421 \\ 0.241 \\ +0.09 \quad \leftarrow \text{(has two decimal places)} \\ \hline 3.4731 \quad \leftarrow \text{(Answer should be reported to two decimal places after rounding off)} \end{array}$$

$$\text{Answer} = 3.47$$

MULTIPLICATION & DIVISION

When numbers are multiplied or divided, the number of significant figures in the answer equals the smallest number of significant figures in any of the original numbers

$$\begin{array}{r} 51.028 \\ \times 1.31 \quad \leftarrow \text{(Three significant figures)} \\ \hline 66.84668 \quad \leftarrow \text{(Answer should have three significant figures after rounding off)} \end{array}$$

$$\text{Answer} = 66.8$$

If $L = 2.331$ cm, $B = 2.1$ cm, then $L+B = ?$

- 4.431 cm
- 4.43 cm
- 4.4 cm
- 4 cm

Dimensional Analysis

Dimensions of a physical quantity are the powers to which units of base quantity are raised. Eg: $[M]^a [L]^b [T]^c [A]^d [K]^e$

APPLICATIONS

checking the correctness of various formulae
Eg: If $Z=A+B$, $[Z]=[A]=[B]$

conversion of one system of unit into another

$$n_1 u_1 = n_2 u_2$$

Eg: $n_1 [M_1^a L_1^b T_1^c] = n_2 [M_2^a L_2^b T_2^c]$

$$n_1 = n_2 \left[\frac{M_2}{M_1} \right]^a \left[\frac{L_2}{L_1} \right]^b \left[\frac{T_2}{T_1} \right]^c$$

Deducing relation among physical quantity

DIMENSIONAL FORMULA

- 1) Pressure = stress = Young's modulus = $M L^{-1} T^{-2}$
- 2) Work = Energy = Torque = $M L^2 T^{-2}$
- 3) Power $P = M L^2 T^{-3}$
- 4) Gravitational constant $G = M^{-1} L^3 T^{-2}$
- 5) Force constant = Spring constant = $M T^{-2}$
- 6) Coefficient of viscosity = $M L^{-1} T^{-1}$
- 7) Latent heat = $L^2 T^{-2}$
- 8) Electric potential = $\frac{P}{I} = M L^2 T^{-3} A^{-1}$
- 9) Resistance = $\sqrt{\frac{H_0}{\epsilon_0}} = M L^2 T^{-3} A^{-2}$
- 10) Capacitance = $M^{-1} L^{-2} T^4 A^2$
- 11) Permittivity $\epsilon_0 = M^{-1} L^{-3} T^4 A^2$
- 12) Angular momentum = Planck's constant = $M^1 L^2 T^{-1}$

$$13) M = k \sqrt{\frac{h c}{G}} \quad L = k \sqrt{\frac{h G}{c^2}} \quad T = k \sqrt{\frac{h G}{c^3}}$$

$$\text{Time period } \frac{L}{R} = RC = \sqrt{LC}$$

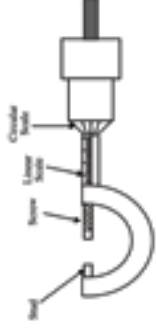
DIMENSIONLESS QUANTITIES

- 1) Strain
- 2) Refractive index
- 3) Relative density
- 4) Plane angle
- 5) Solid angle

In SI Units, the dimensions of $\sqrt{\frac{\epsilon_0}{\mu_0}}$ is:

- a) $A^{-1} T M L^3$
- b) $A T^2 M^{-1} L^{-1}$
- c) $A T^{-3} M L^{3/2}$
- d) $A^2 T^3 M^{-1} L^{-2}$

SCREW GAUGE



$$\text{Pitch} = \frac{\text{Main Scale Reading}}{\text{No. of rotations}}$$

$$\text{Least Count} = \frac{\text{Total no. of divisions on circular scale}}{\text{pitch}}$$

$$\text{Total Reading} = \text{Linear Scale Reading} + \text{circular scale reading} \times \text{least count}$$

The least count of the main scale of a screw gauge is 1mm. The minimum no. of divisions on its circular scale required to measure 5μm diameter of wire is;

- a) 200
- b) 50
- c) 400
- d) 100

COMBINATION OF ERRORS

Operations	Formula Z	Absolute error ΔZ	Relative error ΔZ/Z	Percentage error $100 \times \frac{\Delta Z}{Z}$
Sum	A+B	ΔA+ΔB	$\frac{\Delta A + \Delta B}{A+B}$	$\frac{\Delta A + \Delta B}{A+B} \times 100$
Difference	A-B	ΔA+ΔB	$\frac{\Delta A + \Delta B}{A-B}$	$\frac{\Delta A + \Delta B}{A-B} \times 100$
Multiplication	A×B	ΔAB+BAΔA	$\frac{\Delta A + \Delta B}{A \times B}$	$\left(\frac{\Delta A}{A} + \frac{\Delta B}{B} \right) \times 100$
Division	$\frac{A}{B}$	$\frac{B\Delta A + A\Delta B}{B^2}$	$\frac{\Delta A + \Delta B}{A \times B}$	$\left(\frac{\Delta A}{A} + \frac{\Delta B}{B} \right) \times 100$
Power	A^n	$n A^{n-1} \Delta A$	$\frac{n \Delta A}{A}$	$\frac{n \Delta A}{A} \times 100$
Root	$A^{\frac{1}{n}}$	$\frac{1}{n} A^{\frac{1}{n}-1} \Delta A$	$\frac{1}{n} \frac{\Delta A}{A}$	$\frac{1}{n} \frac{\Delta A}{A} \times 100$

General rule:

If $Z = \frac{A^p B^q}{C^r}$, Then the maximum fractional relative error in Z will be:

$$\frac{\Delta Z}{Z} = p \frac{\Delta A}{A} + q \frac{\Delta B}{B} + r \frac{\Delta C}{C}$$

UNITS & MEASUREMENTS

ERRORS IN MEASUREMENT

Difference between true value & measured value of a quantity

Systematic Errors
Errors which tend to occur only in one direction, either positive or negative

Random Errors
Irregular and random in magnitude & direction

Instrumental

Due to inbuilt defect of measuring instrument

Experimental

Limitations in experimental technique

Personal

Due to individual bias, Lack of proper setting of apparatus

• Least count error is the smallest value that can be measured by instrument (occurs with random & systematic errors)

• Absolute Error :- $\Delta a = a_1 - a_{\text{mean}}, a_{\text{mean}} = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}$

• Relative Error :- $\frac{\Delta a_{\text{mean}}}{a_{\text{mean}}} = \frac{|\Delta a_1| + |\Delta a_2| + |\Delta a_3| + \dots + |\Delta a_n|}{n}$

• Percentage Error :- $\frac{\Delta a_{\text{mean}}}{a_{\text{mean}}} \times 100$

In an experiment four quantities a, b, c and d are measured with percentage error 1%, 2%, 3% and 4% respectively. Quantity P is calculated as shown below. What is the percentage error in P?

$$P = \frac{a^2 b^2}{cd}$$

- (a) 14%
- (b) 10%
- (c) 7%
- (d) 4%